Candidate seat no.:_____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech.(IT) Theory Examination

April-May 2018

MA206.01 Statistical and Numerical Methods

Date: 09.05.2018, Wednesday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

Instructions: 1) The question paper comprises two sections. 2) Section I and II must be attempted in separate answer sheets. 3) Make suitable assumptions and draw neat figures wherever required. 4) Use of scientific calculator is allowed. 5) Candidate must write exam seat number on paper.

SECTION - I

Q-1 **Fill in the blank by choosing the most appropriate answer from the given options.**

(a) The mean and variance of the binomial distribution $b(x;\ n,\ p)$ are _____. **01**

(i) $\mu = p$ & $\sigma^2 = pq$ **(ii)** $\mu = np$ & $\sigma^2 = pq$

(iii) $\mu = np$ & $\sigma^2 = npq$ **(iv)** $\mu = p$ & $\sigma^2 = npq$

(b) If the value of rank correlation coefficient is 1 for given two separate rankings, then both rankings are in _____. **01**

(i) total agreement **(ii)** partial agreement

(iii) total disagreement **(iv)** partial disagreement

(c) Mode of a data set 1, 5, 2, 3, 5, 4 is _____. **01**

(i) 2 **(ii)** 3

(iii) 4 **(iv)** 5

(d) The arithmetic mean of the data set 15, 13, 18, 16, 14, 17, 12 is _____. **01**

(i) 13 **(ii)** 14

(iii) 15 **(iv)** 16

(e) The value of Karl Pearson coefficient lies in the interval _____. **01**

(i) $(-1, 1)$ **(ii)** $[-1, 1)$

(iii) $(-1, 1]$ **(iv)** $[-1, 1]$

(f) If A and B are two event in a sample space S with probability function(P), then $P(A \cup B) =$ _____. **01**

(i) $P(A) + (P(B))^2$ **(ii)** $P(A)P(B)$

(iii) $P(A) + P(B) - P(A \cap B)$ **(iv)** $P(A \cap B)$

(g) If a computer code has no error with probability 0.35, then the probability of an event having at least one error is _____. **01**

(i) 0.65 **(ii)** 0.75

(iii) 0.85 **(iv)** 0.95

Q-2 Attempt any *four*.

(a) Construct cumulative frequency distribution table from following data set: **04**

Lower limit	0.40	2.00	3.60	5.20	6.80	8.40
Upper limit	1.99	3.59	5.19	6.79	8.39	9.99
Number of products	18	13	12	3	3	1

(b) Find Mean for following ungrouped data 2, 5, 12, 10, 20, 15, 18, 26, 25, 36, 34, 30 and 42. Also organize the above data into groups with 5 intervals with left hand convention starting from smallest value and using it evaluates relative frequency (%) of each class. **04**

(c) The rank of some 10 players in two game A and B are given by: **04**

Rank in A	5	2	9	8	1	10	3	4	6	7
Rank in B	10	5	1	3	8	6	2	7	9	4

Calculate the Spearman's rank correlation coefficient and Kendall's rank coefficient.

(d) Find the regression line $y = a + bx$ for the following data: **04**

x	5	2	12	9	15	6	25	16
y	64	87	50	71	44	56	42	60

(e) Define Uniform distribution over the interval (a, b). If a random variable X is uniformly distributed over (a, b), then find its mean. **04**

(f) Assume that the probability of k components malfunctioning within an interval of time t minutes in a system with a large number of components is given by the Poisson pmf with parameter $\lambda = 0.5t$. Compute the probabilities of no components malfunctioning in 20 minutes interval and exactly three number of components malfunctioning in 28 minutes interval. **04**

Q-3 (a) Attempt any *two*.

(i) An investigation on CPU performance gives the frequency distribution of attributes PRP(published relative performance in integer) as in table: **03**

PRP Ranges value ($Lower < PRP \leq Upper$)

Lower limit	0	20	100	180	260	340	420
Upper limit	20	100	180	260	340	420	500
Number of instances	32	120	26	15	6	3	1

Represent the frequency distribution by Histogram.

(ii) The following data pertain to the number of computer job per day and the central processing unit(CPU) time required: **03**

Number of Job	1	2	3	4	5	6
CPU time	3	4.9	7.5	8.99	12	13.2

Draw scatter diagram of above data set.

(iii) A city installs 2000 electrical lamps for street lighting. These lamps have a mean burning life of 1000 hours with standard deviations of 200 hours. What is the probability that a lamp will fail between 900 and 1100 burning hours? You are given that $P(Z \le 0.5) = 0.6915$. **03**

Q-3(b) **Do as directed**

(i) In a 2-week study of the productivity of worker, the following data were obtained on the total number of acceptable pieces of 24 workers produced : **03**

Number of acceptable pieces

25	30	34	35	12	56	52	48
56	18	41	42	13	16	18	40
8	19	23	54	60	58	29	37

Group this data into a distribution having the classes 0-10, 11-20, 21-30, …, 51-60.

(ii) Define Arithmetic mean and write its properties. **03**

OR

Q-3(b) **Do as directed**

(i) Define Mean deviation and standard deviation. **03**

(ii) If two dice be thrown together, in how many ways can they fall? **03**

SECTION – II

Q-4 **Fill in the blank by choosing the most appropriate answer from the given options.**

(a) Consider a data set (x_i, y_i) for i=1 to n, the technique of finding value of y at given x in (x_0, x_n) is called _____. **01**

(i) Extrapolation **(ii)** Interpolation
(iii) Iterative **(iv)** Polynomial equation

(b) Relation between shift operator E and forward operator Δ is given by ____. **01**

(i) $\Delta = E - 1$ **(ii)** $\Delta = E$
(iii) $\Delta = 2E - 1$ **(iv)** $\Delta = E - 2$

(c) _____ is used in Newton backward interpolation formula with step length h. **01**

(i) $(x-x_n)h=u$ **(ii)** $x+x_n=uh$

(iii) $x-x_n=u$ **(iv)** $x-x_n=uh$

(d) One step method for numerical solution of ordinary differential equation is _____. **01**

(i) Euler's method **(ii)** Picard's method

(iii) Runge Kutta method 2nd order **(iv)** Runge Kutta method 4th order

(e) The normal equation for fitting of a straight line is y = a + bx is $\sum y_i =$_____. **01**

(i) $na+b\sum x_i$ **(ii)** $n^2a+b\sum (x_i)^2$

(iii) $na+b\sum (x_i)^2$ **(iv)** $a+b\sum x_i$

(f) The average rate of convergence in Bisection method is_____. **01**

(i) 0.25 **(ii)** 0.5

(iii) 0.75 **(iv)** 1

(g) While evaluating a definite integral by Trapezoidal rule, the accuracy can be increased by taking _____. **01**

(i) larger number of sub-intervals **(ii)** small number of sub-intervals

(iii) odd number of sub-intervals **(iv)** whole interval

Q-5 Attempt any *four*.

(a) A second degree polynomial passes through the points (4, 1), (6, 3), (8, 8) and (10, 16). Using Newton's forward interpolation formula, construct the polynomial and hence compute y (5). **04**

(b) Using Lagrange's interpolation formula and hence find f(3): **04**

x	0	1	2	5
$y=f(x)$	2	3	12	147

(c) Find the equation to the best fitting curve of the form $y=ab^x$ for the given data: **04**

x	1	2	3	4	5
$y=f(x)$	130	150	175	190	240

(d) Evaluate $\sqrt{29}$ to four decimal places by Newton Raphson method. **04**

(e) Obtain Picard's second approximation of the initial value problem $\frac{dy}{dx}=x^2+y^2$ for x = 0.4, given that y (0) = 0. **04**

(f) Evaluate $\int_{2}^{6} \log_{10} x dx$ using Trapezoidal and Simpson's 1/3rd rule by taking seven grid points. **04**

Q-6 (a) Attempt any *two.*

(i) Form divided difference table for the given data: **03**

x	40	50	60	70	80	90
$y = f(x)$	204	224	246	270	296	324

(ii) The following table gives the population (thousand) of a town during the last six censuses. Estimate, using Newton's interpolation formula, the increase in population from 1961 to 1966. **03**

Year	1911	1921	1931	1941	1951	1961
population	12	15	20	27	39	52

(iii) Find a real root of $x^3 - 4x - 9 = 0$ using bisection method up to fourth approximation. **03**

Q-6(b) Do as directed

(i) Use second order Runge-Kutta method with h = 0.1 , find y_1 and y_2 for $\frac{dy}{dx} = x - y^2,\ y(0) = 1$. **03**

(ii) Find the real root of $\cos x - 3x + 5 = 0$ using False position method correct to three decimal places. **03**

OR

Q-6(b) Do as directed

(i) An experimental data about force and velocity for an object suspended in a wind tunnel is given as **03**

Velocity(m/s)	10	20	30	40	50	60	70	80
Force (N)	24	68	378	552	608	1218	831	1452

Use least-squares regression to determine the coefficients *a* and *b* in the function y = *a* + *bx* that best fits the data.

(ii) Use Euler's method to find y at x = 0.2 from the differential equation $\frac{dy}{dx} = \frac{y}{2}; y(0) = 1$ using step size h = 0. 1. **03**

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